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### REMOVABLE CLIP

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#### Abstract

A removable clip for fastening together a first piece adjacent to a second piece to form a portion of a crate. The removable clip includes a body. The body includes a first arm, a second arm coupled to the first arm, a third arm coupled to the second arm, a fourth arm coupled to the third arm, a fifth arm, and a sixth arm provided on the fifth arm between a first distal end and a second distal end. A first channel receives the first piece and a second channel receives the second piece.

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## Background/Summary

CROSS REFERENCE [0001] This application is a continuation-in-part of International Patent Application No. PCT/US2023/081595 filed on Nov. 29, 2023, all of which is incorporated herein by reference in its entirety.

### BACKGROUND

[0002] Crates, also known as containers or boxes, have been used for centuries to move materials or goods from one location to another. When carrying heavy loads, boxes are often made from increasingly stronger materials such as wood or metal. Often the sides of the crate that define a protected or enclosed volume are secured to one another by means of nails, screws, or tightening bands requiring specific tools to remove. Optionally, these crates can be secured to wooden pallets, allowing the use of machines to move the crate.

[0003] Traditionally, crates are designed for single use and disposed of after the transported materials or goods have reached their destination. However, due to financial, environmental and/or safety reasons, reusable collapsible crates have been developed and are increasingly used for transporting materials and goods.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0004] In the drawings:

[0005] FIG. 1 is a perspective view of an assembled reusable crate with a plurality of clips.

[0006] FIG. 2 is a perspective view of a clip of the plurality of clips of FIG. 1 according to aspects of the present disclosure.

[0007] FIG. 3 is a cross-sectional view of the clip of FIG. 2 according to aspects of the present disclosure.

[0008] FIG. 4 is a cross-sectional view of the clip of FIG. 2 and FIG. 3 coupling adjacent pieces according to aspects of the present disclosure.

[0009] FIG. 5 is a variation of the cross-sectional view of FIG. 3 according to aspects of the present disclosure.

[0010] FIG. 6 is a variation of the perspective view of FIG. 2 according to aspects of the present disclosure.

[0011] FIG. 7 is a cross-sectional view of the clip of FIG. 6 according to aspects of the present disclosure.

[0012] FIG. 8 is a cross-sectional view of the clip of FIG. 6 and FIG. 7 coupling adjacent pieces according to aspects of the present disclosure.

### DETAILED DESCRIPTION

[0013] Collapsible or reusable crates provide the greatest benefit when designed to be easily assembled to form small or large containers by a single person without the use of nails, a hammer, or other traditional tools and equipment traditionally associated with assembling crates.

[0014] The present disclosure relates to a removable clip for a knock-down structure, and more particularly, to a removable clip for use to secure adjacent pieces of a collapsible container to one another. As disclosed herein, a clip releasably couples two adjacent pieces of a reusable crate. That is, side walls of the reusable crate are connected to one other and to a base using a plurality of removable clips. A lid can then be coupled to one or more of the side walls. The base can be formed with or coupled to a skid, such that, when assembled, the reusable crate can be handled by a machine, such as a forklift.

[0015] The removable clips are tool-free. As used herein, the term “tool-free” defines the assembly,

application, mounting, fastening, or construction of an object or portions of the object without the use of tools, such as, but not limited to a hammer, screw driver, drill, or pry bar.

[0016] Since the removable clips are tool-free, the reusable crate can be assembled by a single person using the removable clips without any tools. Further the single person can also disassemble the reusable crate by the removal of the clips without the use of tools.

[0017] As used herein, the term “piece” is anything that can be used to form at least a portion of a container. For example, a piece can be lumber approximately 2 inches by 4 inches as sawed and usually 1.5 inches by 3.5 inches when dressed. By way of further non-limiting example, a piece can be lumber approximately 1 inch by 2 inches, 2 inches by 6 inches, 4 inches by 4 inches, or 2 inches by 8 inches, as sawed. Further, a piece can include or be a panel, a plywood sheet, or a portion of a plywood sheet having a thickness, for example, of 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 1.125, or 1.25 inches. While illustrated as lumber, a piece can have any dimensions and be made entirely from or include portions of that are plastic, wood, or metal.

[0018] Each removable clip generally has a G-shape formed by at least five arms. The five arms, along with a sixth arm or spur arm, contact the two adjacent pieces at six surfaces. A finger grip defined by a portion of one of the arms allows for easy removal of the removable clip. A radiused or rounded portion of the sixth arm eases fastening of the removable clip to the two pieces.

[0019] All directional references (e.g., proximal, distal, upper, lower, upward, downward, left, right, lateral, front, back, top, bottom, above, below, vertical, horizontal, etc.) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of aspects of the disclosure described herein. Connection references (e.g., attached, coupled, secured, fastened, and connected,) are to be construed broadly and can include intermediate members between a collection of elements and relative movement between elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and in fixed relation to one another. The exemplary drawings are for purposes of illustration only and the dimensions, positions, order and relative sizes reflected in the drawings attached hereto can vary.

[0020] While “a set of” or “a plurality of” various elements will be described, it will be understood that “a set” or “a plurality” can include any number of the respective elements, including only one element.

[0021] FIG. 1 illustrates a perspective view of a collapsible or reusable crate assembly **10** in an assembled state. The reusable crate assembly **10** includes a set of pieces **12** and a plurality of removable clips **14**. The set of pieces **12** are illustrated, by way of example, as a base piece **20**, a lid piece **22**, a first side piece **24**, a second side piece **26**, a third side piece **28**, and a fourth side piece **30**. The first side piece **24**, the second side piece **26**, the third side piece **28**, and the fourth side piece **30** can extend between the base piece **20** and the lid piece **22**. While illustrated as having four side pieces, any number of side pieces are contemplated, such as, but not limited to, three, five, or six.

[0022] Each piece of the set of pieces **16** can include a frame **32**. The frame **32** can define a perimeter of the piece. Optionally, one or more cross members **34** can couple two or more portions of the frame **32**. The frame **32**, as illustrated by way of example, can be defined by a set of frame pieces coupled together by one or more fasteners **36**. Other fasteners **38** can couple the one or more cross members **34** to portions of the frame **32** and/or couple together adjacent portions of the frame **32**.

[0023] Each removable clip of the plurality of removable clips **14** couples together two adjacent pieces. While two removable clips of the plurality of removable clips **14** are illustrated as coupling adjacent pieces of the set of pieces **16**, any number of removable clips are contemplated. Depending on a size of the reusable crate assembly **10** a single removable clip could be utilized to couple together two adjacent pieces.

[0024] As illustrated, by way of example, a first removable clip **14a** and a second removable clip

**14b** of the plurality of removable clips **14** fasten the first side piece **24** and the second side piece **26**. By way of further non-limiting example, a third removable clip **14c** and a fourth removable clip **14d** couple the second side piece **26** to the base piece **20**.

[0025] The base piece **20** is illustrated, by way of example, as a pallet having receiving portions **40** between vertical supports **42**, where the receiving portions **40** can receive one or more machines such as, but not limited to one or more of a forklift, a pallet jack, a front loader, a jacking device, or an erect crane.

[0026] The set of pieces **16** can be formed from a variety of materials, including, but not limited to one or more of wood, plastic, metal, natural fibers, composite, or recycled material. By way of non-limiting example, the wood can be one or more of hardwoods like hickory, mahogany, maple, oak, teak, and walnut, or softwoods like various types of pine. Each piece of the set of pieces **16** can include different materials or different combinations or ratios of materials.

[0027] While illustrated as open, one or more pieces of the set of pieces **16** can include one or more sheets of material that couple to the frame **32**, one or more cross members **34**, or both the frame **32** and the one or more cross member **34**. The sheets of material can include cut-outs, recesses, or flex in a way that the plurality of removable clips **14** can still couple the frames **32** of adjacent pieces of the set of pieces **16**.

[0028] The set of pieces **16** can include lumber approximately 2 inches by 4 inches as sawed and usually 1.5 inches by 3.5 inches when dressed. By way of further non-limiting example, the set of pieces **16** can include lumber approximately 1 inch by 2 inches, 2 inches by 6 inches, 4 inches by 4 inches, 2 inches by 8 inches, 2 inches by 10 inches, or 2 inches by 12 inches as sawed. Further, the set of pieces **16** can include a panel, a plywood sheet, or a portion of a plywood sheet having a thickness, for example, of 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 1.125, or 1.25 inches. As illustrated, by way of example, the dimensions each piece of the set of pieces **16** can vary. Therefore, the removable clip **14** can have a varying profile with varying arm lengths that depend on dimensions of the pieces used to form the reusable crate assembly **10**.

[0029] FIG. **2** is a perspective view of the first removable clip **14a** of the plurality of removable clips **14**. The first removable clip **14a** is similar to the other removable clips of the plurality of removable clips **14** and hereinafter will be referred to as the removable clip **14a**.

[0030] The removable clip **14a** has a body **50**. The body **50** includes a first arm **52**, a second arm **54** extending from to the first arm **52**, a third arm **56** extending from the second arm **54**, a fourth arm **58** extending from the third arm **56**, a fifth arm **60** extending from the fourth arm **58**, and a spur arm or a sixth arm **62** provided on the fifth arm **60**. The fifth arm **60** includes a first distal end **64** where it intersects with the fourth arm **58**, and a second distal end **66**, which is a free end. The sixth arm **62** extends from the fifth arm **60** at a portion that is between the first distal end **64** and the second distal end **66**. The first arm **52**, the second arm **54**, the third arm **56**, the fourth arm **58**, and the fifth arm **60** form a G-shaped clip.

[0031] Optionally, a set of spurs **68** can be provided extending from an interior side of the second arm **54**. While illustrated as having two spurs in the set of spurs **68**, the set of spurs **68** can include any number of spurs. Further, while illustrated as extending from the second arm **54**, it is contemplated that the first arm **52**, third arm **56**, fourth arm **58**, fifth arm **60**, and/or sixth arm **62** could include one or more spurs.

[0032] A first corner **70** is defined in the body **50** where the first arm **52** extends from the second arm **54**. That is, the first corner **70** is located at the intersection of the first arm **52** and the second arm **54**. A second corner **72** is defined in the body **50** where the second arm **54** extends from the third arm **56**. That is, the second corner **72** is located at the intersection of the second arm **54** and the third arm **56**. A third corner **74** is defined in the body **50** where the third arm **56** extends from the fourth arm **58**. That is, the third corner **74** is located at the intersection of the third arm **56** and the fourth arm **58**. A fourth corner **76** is defined in the body **50** where the fourth arm **58** extends from the fifth arm **60**. That is, the third corner **74** is located at the intersection of the fourth arm **58**

and the fifth arm **60**.

[0033] The removeable clip **14a** can be unitary body or unitarily formed. That is, the body **50** can be a unitary monolithic body. It is contemplated that the body **50** can be formed by molded plastic. Further, it is contemplated that the body **50** can include at least one polymer. Additionally, or alternatively, the body **50** of the removeable clip **14a** can include one or more of metal (e.g., steel, aluminum) or natural fiber where natural fiber can be produced by geological processes or from the bodies of plants or animals. It is contemplated that the features of the body **50** may be formed from a singular piece of metal such a wire, as further illustrated in FIG. 5.

[0034] It is further contemplated that each removable clip of the plurality of removable clips **14** (FIG. 1) can be made from the same material. Alternatively, one or more removable clip of the plurality of removable clips **14** (FIG. 1) can include different material(s) or different proportions of materials.

[0035] Optionally, the body **50** can include a window **78**. While illustrated as a single window **78** in the fourth arm **58**, any number of windows located in any one or more arms is contemplated. The window **78** can be a recessed portion or extend through the body **50**. It is further contemplated that one or more windows can extend around corners. That is, the window **78** can extend from at least one arm to another.

[0036] FIG. 3 is a cross sectional view of the removable clip **14a** further illustrating the body **50**. The first corner **70** defines a first angle **80**, the second corner **72** defines a second angle **82**, the third corner **74** defines a third angle **84**, and the fourth corner **76** defines a fourth angle **86**. The first angle **80**, the second angle **82**, the third angle **84**, and the fourth angle **86** are between 10 degrees and 110 degrees. The third angle **84** can be in a range from 20 degrees to less than 90 degrees. By way of example, the third angle **84** can be in a range from 70 degrees to less than 90 degrees.

[0037] A first surface **90** is defined by an interior portion of the first arm **52**. A length, illustrated as a first arm length **92**, of the first arm **52** is measured from a first free end **94** to the first corner **70**. By way of non-limiting example, the first arm length **92** can be measured along the first surface **90**.

[0038] A second surface **96** is defined by an interior portion of the second arm **54**. A length, illustrated as a second arm length **98**, of the second arm **54** is measured from the first corner **70** to the second corner **72**. By way of non-limiting example, the second arm length **98** can be measured along the second surface **96**.

[0039] The second arm length **98** can be determined by a first thickness of a first piece of the two adjacent pieces to be held together by the removeable clip **14a**. By way of non-limiting example, the second arm length **98** can be within 5% or less of the first thickness. That is, the second arm length **98** can be within 5% or less of a finished or dressed first thickness of the first piece that can be, for example, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 1.0, 1.125, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5, 2.75, 3.0, 3.25, or 3.5 inches.

[0040] As illustrated, by way of non-limiting example, the first arm length **92** and the second arm length **98** can be equal. As used herein “equal” means that the percent difference between the two values is 5% or less. Alternatively, in a different and non-limiting example, the first arm length **92** can be greater than the second arm length **98**. That is, the first arm length **92** can be more than 5% greater than the second arm length **98**. In yet another different and non-limiting example, the second arm length **98** can be more than 5% greater than the first arm length **92**.

[0041] The set of spurs **68** are illustrated, by way of example, as located between the first corner **70** and the second corner **72**. Each spur of the set of spurs **68** can include a spur height **100** and a spur width **102**. The spur height **100** is measure from the surface from which the spur extends, illustrated as the second surface **96** to a spur peak **104** or portion of the spur extending farthest from the second surface **96**. The spur height **100** can be in a range from 2% to 30% of the second arm length **98**. For example, the spur height **100** can be in a range from 5% to 20% of the second arm length **98**. In another non-limiting example, the spur height **100** can be less than 15% of the second arm length **98**.

[0042] The spur width **102** can be measured along the second surface **96**. The spur width **102** is the distance between the two points at which the spur intercepts the second surface **96**.

[0043] The spur width **102** can be in a range from 2% to 50% of the second arm length **98**. For example, the spur width **102** can be in a range from 10% to 30% of the second arm length **98**. It is further contemplated that the spur width **102** can be in a range from 25% to 150% of the spur height **100**. For example, the spur width **102** can be in a range from 40% to 60% of the spur height **100**.

[0044] A third surface **106** is defined by an interior portion of the third arm **56**. A length, illustrated as a third arm length **108**, of the third arm **56** is measured from the second corner **72** to the third corner **74**. By way of non-limiting example, the third arm length **108** can be measured along the third surface **106**.

[0045] As illustrated, by way of non-limiting example, the third arm length **108** is greater than the first arm length **92** and the second arm length **98**. It is contemplated that, for example, the third arm length **108** can be two to ten times longer than the first arm length **92**.

[0046] Alternatively, in a different and non-limiting example, the third arm length **108** can be equal to the first arm length **92** or the second arm length **98**. In yet another different and non-limiting example, the third arm length **108** can be more than 5% less than the first arm length **92**, the second arm length **98**, or both the first arm length **92** and the second arm length **98**.

[0047] The third arm length **108** can be determined by a first width of the first piece of the two adjacent pieces to be held together by the removeable clip **14a**. By way of non-limiting example, the third arm length **108** can be within 5% or less of the first width of the first piece. That is, the third arm length **108** can be within 5% or less of a finished or dressed first width of the first piece that can be, for example, 0.25, 0.5, 0.75, 1.0, 1.5, 3.5, 5.5, 7.25, 9.25, or 11.25 inches.

[0048] A fourth surface **110** is defined by an interior portion of the fourth arm **58**. A length, illustrated as a fourth arm length **112**, of the fourth arm **58** is measured from the third corner **74** to the fourth corner **76**. By way of non-limiting example, the fourth arm length **112** can be measured along the fourth surface **110**.

[0049] As illustrated, by way of non-limiting example, the fourth arm length **112** is greater than the third arm length **108**, the second arm length **98**, and the first arm length **92**. Although this need not be the case. It is contemplated that, for example, the fourth arm length **112** can be two to ten times longer than the first arm length **92**.

[0050] The fourth arm length **112** can be determined by a second width of a second piece of the two adjacent pieces to be held together by the removeable clip **14a**. By way of non-limiting example, the fourth arm length **112** can be within 5% or less of the second width of the second piece plus the first thickness of the first piece. That is, fourth arm length **112** can be within 5% or less of a finished or dressed second width of the second piece that can be, for example, 0.25, 0.5, 0.75, 1.0, 1.5, 3.5, 5.5, 7.25, 9.25, or 11.25 inches added to the first thickness of the first piece that can be, for example, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 1.0, 1.125, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5, 2.75, 3.0, 3.25, or 3.5 inches.

[0051] A fifth surface **114** is defined by an interior portion of the fifth arm **60**. A length, illustrated as a fifth arm length **116**, of the fifth arm **60** is measured from the fourth corner **76** to the second distal end **66** of the fifth arm **60**. By way of non-limiting example, the fifth arm length **116** can be measured along the fifth surface **114**.

[0052] By way of non-limiting example, the fifth arm length **116** can be between 50% and 250% of the first arm length **92**. As illustrated, by way of non-limiting example, the fifth arm length **116** can be in a range from 100% to 250% of the first arm length **92**. Further, the fifth arm length **116** is also illustrated, by way of example, as less than the third arm length **108**.

[0053] The fifth arm length **116** can be determined by a second thickness of the second piece of the two adjacent pieces to be held together by the removeable clip **14a**. By way of non-limiting example, the fifth arm length **116** can be within 5% or less of the second thickness. That is, fifth

arm length **116** can be within 5% or less of a finished or dressed second thickness of the second piece that can be, for example, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 1.0, 1.125, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5, 2.75, 3.0, 3.25, or 3.5 inches.

[0054] A sixth surface or spur surface **120** is defined by an interior portion of the sixth arm **62**. A length, illustrated as a sixth arm length **122**, of the sixth arm **62** is measured from the fifth surface **114** to a free end **124**. The sixth arm length **122** is greater than the spur height **100**.

[0055] The sixth arm length **122** is in a range of 5% to 100% of the first arm length **92**. For example, the sixth arm length **122** can be in a range of 5% to 75%, 10% to 50%, 10% to 40%, 15% to 80% or 20% to 70% of the first arm length **92**.

[0056] Further, it is contemplated that the sixth arm length **122** is in a range of 5% to 100% of the fifth arm length **116**. For example, the sixth arm length **122** can be in a range of 5% to 75%, 10% to 50%, 20% to 80%, or 40% to 60% of the fifth arm length **116**.

[0057] Over the sixth arm length **122**, the spur surface **120** can include a curved portion **126**. The curved portion **126** of the spur surface **120** has a radius of curvature greater than zero. The curved portion **126** defines at least a portion of the free end **124**.

[0058] An inset length **128** is defined as the length between the second distal end **66** of the fifth arm **60** and an outer portion **130** of the sixth arm **62**.

[0059] A finger grip **132** is defined by a portion of the fifth arm **60** between the outer portion **130** of the sixth arm **62** and the second distal end **66**. The finger grip **132** can include a portion adjacent the outer portion **130**. The finger grip **132** can have a radius of curvature greater than zero. The finger grip **132** has a length illustrated as the inset length **128**.

[0060] A first channel **134** is defined by the first arm **52**, the second arm **54**, and the third arm **56**. A second channel **136** is defined by the fourth arm **58**, the fifth arm **60**, and the sixth arm **62**.

[0061] FIG. 4 further illustrates the removable clip **14a** in a mounted position or assembled with a first piece **140** and a second piece **142**. That is, the removable clip **14a** is coupling adjacent frames **32** (FIG. 1) illustrated as a first piece **140** and a second piece **142**, adjacent to the first piece **140**. The first channel **134** receives the first piece **140**. In other words, when the removable clip **14a** is coupling the first piece **140** and the second piece **142**, the first surface **90**, the second surface **96** and the third surface **106** are adjacent portions the first piece **140**. As used herein, the term “adjacent” means that a surface of one object is facing, next to, or in contact with a surface of a second object.

[0062] The second channel **136** receives the second piece **142**. In other words, when the removable clip **14a** is coupling the first piece **140** and the second piece **142**, the fourth surface **110**, the fifth surface **114**, and the spur surface **120** are adjacent, facing, and/or in contact with portions the second piece **142**.

[0063] The first piece **140** includes a piece distal end **144** that is received by the first channel **134**. When the removable clip **14a** is fastened to the first piece **140** and the second piece **142**, the set of spurs **68** can be in contact with the piece distal end **144**. Alternatively, the set of spurs **68** can pierce, detent, or extend into the first piece **140** past the piece distal end **144**. Additionally, or alternatively, the piece distal end **144** contacts the second surface **96**.

[0064] A third wall thickness **146** can be measured from the third surface **106** to an outer surface **148** of the third arm **56**. Similarly, a fourth wall thickness **150** can be measured from the fourth surface **110** to an outer surface **152** of the fourth arm **58**. A third corner thickness **156** can be measured from the intersection of the third surface **106** and the fourth surface **110** to an outer surface **158** of the third corner **74**. The third corner thickness **156** can be greater than the third wall thickness **146** and the fourth wall thickness **150**. That is, the body **50** can include thickened portions at the third corner **74**. Further, as illustrated by way of example, the body **50** can include thickened portions at the first corner **70**, the second corner **72**, the third corner **74**, and the fourth corner **76**. It is contemplated that any or all of the first corner **70**, the second corner **72**, the third corner **74**, or the fourth corner **76** can include additional material or thickened portions. The

thickened portions can be formed as part of the monolithic body **50**. Alternatively, the additional material or thickened portions can be coupled to the body **50** at any location.

[0065] Referring to FIG. 1-FIG. 4, in operation, to assemble the reusable crate assembly **10**, the base piece **20** can receive one or more side pieces. Building or assembly of the reusable crate assembly **10** can be completed by a single person without tools. The side pieces can be fastened to the base piece **20** using one or more removable clips from the plurality of removable clips **14**. By way of non-limiting example, a portion of the frame **32** of the base piece **20** can be received by the first channel **134** of the removable clip **14c**. Force provided by the individual putting the reusable crate assembly **10** together is applied to the body **50** including by way of non-limiting example, the third arm **56**, the fourth arm **58** or a combination thereof and brings a portion of the frame **32** of the second side piece **26** into the second channel **136** of the removable clip **14c**. The curved portion **126** of the sixth arm **62** eases the entrance of the portion of the frame **32** of the second side piece **26** into the second channel **136**. The curved portion **126** makes it easier for the removable clip **14a** at the sixth arm **62** to slid over sharp corners of the second piece **142**.

[0066] The removable clip **14c**, having a G-shape, wraps around the base piece **20** and the second side piece **26** when fastened. That is, the removable clip **14c** contacts six total surfaces. The first channel **134** contacts three surfaces of the base piece **20**, where each of the three contacted surfaces is in a different plane than the other two. As used herein, the term “plane” is a two-dimensional space or flat surface that extends indefinitely. In other words, the three surfaces contacted by the removable clip **14c** can include two parallel planes intersected by a third plane. Additionally, the second channel **136** contacts three surfaces of the second side piece **26**, where each of the three contacted surfaces of the second side piece **26** define a different plane than the other two.

[0067] The removable clip **14c** can be made of a material such a polymer or metal that has both strength benefits and provides a desired amount of flexibility. The flexibility of the removable clip **14c** can also ease the installation of the wrap-around removable clip **14c**, while the strength of the material provides a secure and durable fastener between the base piece **20** and the second side piece **26**.

[0068] Optionally, additional removable clips, such as removable clip **14d** can further secure the second side piece **26** to the base piece **20**.

[0069] The first side piece **24** can be located with respect to the base piece **20** and the second side piece **26**. The first channel **134** of the removable clip **14a** can be received by a portion of the frame **32** of the second side piece **26**. A portion of the frame **32** of the first side piece **24** can be received by the second channel **136** of the removable clip **14a**. Optionally, additional removable clips, such as removable clip **14b** can further secure the first side piece **24** to the second side piece **26**.

[0070] Similar to the second side piece **26**, the fourth side piece **30** can couple to the base piece **20** and the first side piece **24** using one or more removable clip from the plurality of removable clips **14**.

[0071] The third side piece **28** can couple to the second side piece **26** using one or more removable clip from the plurality of removable clips **14**. Additionally, the third side piece **28** can couple to the fourth side piece **30** using one or more removable clip from the plurality of removable clips **14**.

[0072] The lid piece **22** can couple to the second side piece **26** using one or more removable clip from the plurality of removable clips **14**. Additionally, the lid piece **22** can couple to the fourth side piece **30** using one or more removable clip from the plurality of removable clips **14**.

[0073] To disassemble the reusable crate assembly **10**, an individual can use their hand or fingers to apply force to the finger grip **132** to remove each removable clip of the plurality of removable clips **14**. The curved portion **126** of the sixth arm **62** eases the removal of the plurality of removable clips **14** when force is applied to the finger grips **132**. Therefore, disassembly of the reusable crate assembly **10** can be completed by an individual without the use of tools.

[0074] FIG. 5 is a cross-sectional view of a removable clip **214** formed from one or more pieces of

metal such as a wire. The removable clip **214** is similar to the removable clip **14a** of FIG. 3; therefore, like parts will be identified with like numerals increased by 200, with it being understood that the description of the like parts of the removable clip **14a** applies to the removable clip **214**, except where noted.

[0075] The removable clip **214** has a body **250**. The body **250** includes a first arm **252**, a second arm **254** extending from to the first arm **252**, a third arm **256** extending from the second arm **254**, a fourth arm **258** extending from the third arm **256**, a fifth arm **260** extending from the fourth arm **258**, and a spur arm or sixth arm **262** extending from the fifth arm **60**.

[0076] The sixth arm **262**, is illustrated the wire looped on itself. The sixth arm **262** extends from the fifth arm **260** at a portion that is between a first distal end **264** and a second distal end **266** of the fifth arm **260**. The first arm **252**, the second arm **254**, the third arm **256**, the fourth arm **258**, and the fifth arm **260** form a G-shaped clip.

[0077] Aspects of the present disclosure provide a variety of benefits including that the removable clips are a tool free fastener. That is, without tools, a single individual can assemble the reusable crate assembly using the removable clips to couple adjacent pieces of the crate together. The reusable crate assembly can be shipped and a single individual who receives the reusable crate assembly can disassemble the crate, again, without the use of tools.

[0078] Another benefit is the wrap-around design in which the removable clip contacts six surfaces when applied. The wrap-around design provides increase security in the fastening of the pieces and the fastening of the removable clip to the pieces.

[0079] Another benefit is a lower cost to manufacture. The removable clip can be reused as it is not screwed, hammered, or otherwise applied to the reusable crate with a tool or fastener. Further, the removable clips can be made or cast from a polymer material, which are traditionally less expensive than clips made of metal. Polymers or plastics can provide an increase in flexibility at predetermined locations.

[0080] Yet another benefit is the set of spikes on the inside edge or the second surface of the second arm. The set of spurs provide pressure to the first piece, securing the first piece against the second piece in the clip.

[0081] The radius edge or curved portion of the sixth arm makes it easier for a hand to snap or push the removable clip into place. The radius edge or curved portion also makes it easier to release the removable clip via the finger grip.

[0082] Increased structure illustrated as the thickened portions at the first corner, the second corner, the third corner, the fourth corner, and/or the fifth corner provide additional strength and rigidity.

[0083] The finger grip allows a human hand to easily release or remove the removable clip with no tools.

[0084] The third angle is tapered; that is, the third angle is less than 90 degrees. The third angle allows for a tighter fit; pressing the fourth arm and the third arm against the first piece and the second piece.

[0085] FIG. 6 illustrates a removable clip **314**. The removable clip **314** is similar to the removable clip **14a** of FIG. 2; therefore, like parts will be identified with like numerals increased by three hundred (300), with it being understood that the description of the like parts of the removable clip **14a** applies to the removable clip **314**, except where noted.

[0086] The removable clip **314** has a body **350**. The body **350** includes a first arm **352**, a second arm **354** extending from to the first arm **352**, a third arm **356** extending from the second arm **354**, a fourth arm **358** extending from the third arm **356**, a fifth arm **360** extending from the fourth arm **358**, and a spur arm or a sixth arm **362** provided on the fifth arm **360**. The fifth arm **360** includes a first distal end **364** where it intersects with the fourth arm **358**, and a second distal end **366**, which is a free end. The sixth arm **362** extends from the fifth arm **360** at a portion that is between the first distal end **364** and the second distal end **366**. The first arm **352**, the second arm **354**, the third arm **356**, the fourth arm **358**, and the fifth arm **360** form a G-shaped clip.

[0087] A first subset of spurs **369a** of a set of spurs **369** extend from an interior surface **371** of the second arm **354**. A second subset of spurs **369b** of the set of spurs **369** extend from an interior surface **373** of the fifth arm **360**. The second subset of spurs **369b** can be located between the fourth arm **358** and the sixth arm **362**. The set of spurs **369** can include any number of spurs or subsets of spurs. Further, while illustrated as extending from the second arm **354** and the fifth arm **360**, it is contemplated that the first arm **352**, third arm **356**, fourth arm **358**, and/or sixth arm **362** could include one or more spurs.

[0088] A spur length **363** can be measured in a direction **L** that extends perpendicular to a front face **365** of the body **350** and a rear face **367**, opposite of the front face **365** of the body **350**.

[0089] While illustrated as having a uniform size and shape, it is contemplated that that two or more spurs of the set of spurs **369** can have different sizes. For example, two or more spurs of the set of spurs **369** can have different heights, lengths, and/or widths as further described in FIG. 7. While illustrated as triangular prisms, it will be understood that this is by way of non-limiting example only and it is contemplated that any number of the set of spurs **369** can have any suitable shape, geometry, size, surface textures, or profile. Further still, by way of further non-limiting example, subsets of the set of spurs **369** can have different shapes, geometries, sizes, surface textures, or profiles. By way of additional non-limiting examples, one or more spurs of the set of spurs **369** can be a triangular pyramid, rectangular pyramid, or any regular or irregular pyramid. It is further contemplated that at least one spur of the set of spurs can have a cross section that is a rectangle, triangle, semi-circle, regular polygon, or irregular polygon.

[0090] A first corner **370** is defined in the body **350** where the first arm **352** extends from the second arm **354**. That is, the first corner **370** is located at the intersection of the first arm **352** and the second arm **354**. A second corner **372** is defined in the body **350** where the second arm **354** extends from the third arm **356**. That is, the second corner **372** is located at the intersection of the second arm **354** and the third arm **356**. A third corner **374** is defined in the body **350** where the third arm **356** extends from the fourth arm **358**. That is, the third corner **374** is located at the intersection of the third arm **356** and the fourth arm **358**. A fourth corner **376** is defined in the body **350** where the fourth arm **358** extends from the fifth arm **360**. That is, the third corner **374** is located at the intersection of the fourth arm **358** and the fifth arm **360**.

[0091] The removeable clip **314** can be unitary body or unitarily formed. That is, the body **350** can be a unitary monolithic body. It is contemplated that the body **350** can be formed by molded plastic. Further, it is contemplated that the body **350** can include at least one polymer. Additionally, or alternatively, the body **350** of the removeable clip **314** can include one or more of metal (e.g., steel, aluminum) or natural fiber where natural fiber can be produced by geological processes or from the bodies of plants or animals.

[0092] Optionally, the body **350** can include a feature **378**. While illustrated as a single feature **378** in the fourth arm **358**, it is contemplated that any number of features can be suitably located in any one or more arms. The feature **378** can be a recessed portion or can extend through the body **350** to create one or more windows. Alternatively, in another non-limiting example, the feature can be an embossment or raised portion of the body **350**. By way of non-limiting example, the feature **378** can be a raised logo. In yet another different and non-limiting example, the feature **378** can be a thickened portion of the body **350**. In yet a still further non-limiting example, the feature **378** can include a differing surface texture from a remainder of the surrounding body **350**. It is further contemplated that one or more features can extend around corners. That is, the feature **378** can extend from at least one arm to another.

[0093] FIG. 7 is a cross sectional view of the removable clip **314** further illustrating the body **350**. The first corner **370** defines a first angle **380**, the second corner **372** defines a second angle **382**, the third corner **374** defines a third angle **384**, and the fourth corner **376** defines a fourth angle **386**. The first angle **380**, the second angle **382**, the third angle **384**, and the fourth angle **386** are between 10 degrees and 110 degrees. For example, the first angle **380**, the second angle **382**, and the fourth

angle **386** can be in a range from 80 degrees to 100 degrees. By way of further non-limiting example, the first angle **380**, the second angle **382**, and the fourth angle **386** can be in a range from 85 degrees to 95 degrees.

[0094] The third angle **384** can be in a range from 20 degrees to less than 90 degrees. By way of example, the third angle **384** can be in a range from 70 degrees to less than 90 degrees.

[0095] It is contemplated that the third angle **384** can be less than one or more of the first angle **380**, the second angle **382**, and the fourth angle **386**.

[0096] A first surface **390** is defined by an interior portion of the first arm **352**. A length, illustrated as a first arm length **392**, of the first arm **352** is measured from a first free end **394** to the first corner **370**. By way of non-limiting example, the first arm length **392** can be measured along the first surface **390**.

[0097] A second surface **396** is defined by an interior portion of the second arm **354**. A length, illustrated as a second arm length **398**, of the second arm **354** is measured from the first corner **370** to the second corner **372**. By way of non-limiting example, the second arm length **398** can be measured along the second surface **396**.

[0098] The second arm length **398** can be determined by a first thickness of a first piece of the two adjacent pieces to be held together by the removeable clip **314**. By way of non-limiting example, the second arm length **398** can be within 10% or less of the first thickness. That is, the second arm length **398** can be within 10% or less of a finished or dressed first thickness of the first piece that can be, for example, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 1.0, 1.125, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5, 2.75, 3.0, 3.25, or 3.5 inches. For example, the second arm length **398** can be within 5% or less of the first thickness. That is, the second arm length **398** can be within 5% or less of a finished or dressed first thickness of the first piece that can be, for example, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 1.0, 1.125, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5, 2.75, 3.0, 3.25, or 3.5 inches.

[0099] As illustrated, by way of non-limiting example, the first arm length **392** can be less than the second arm length **398**. Alternatively, in a different and non-limiting example, the first arm length **392** can be greater than the second arm length **398**.

[0100] The first subset of spurs **369a** are illustrated, by way of example, as located between the first corner **370** and the second corner **372**. Each spur of the first subset of spurs **369a** can include a spur height **400** and a spur width **402**. The spur height **400** is measure from the surface from which the spur extends, illustrated as the second surface **396** to a spur peak **404** or portion of the spur extending farthest from the second surface **396**. The spur height **400** can be in a range from 2% to 30% of the second arm length **398**. For example, the spur height **400** can be in a range from 5% to 20% of the second arm length **398**. In another non-limiting example, the spur height **400** can be in a range from 10% to 15% of the second arm length **398**.

[0101] The spur width **402** can be measured along the second surface **396**. The spur width **402** is the distance between the two points at which the spur intercepts the second surface **396**.

[0102] The spur width **402** can be in a range from 2% to 50% of the second arm length **398**. For example, the spur width **402** can be in a range from 10% to 30% of the second arm length **398**. In another non-limiting example, the spur width **402** can be in a range from 15% to 20% of the second arm length **398**. It is further contemplated that the spur width **402** can be in a range from 25% to 150% of the spur height **400**. For example, the spur width **402** can be in a range from 50% to 150% of the spur height **400**. In another non-limiting example, the spur width **402** can be in a range from 110% to 140% of the spur height **400**.

[0103] The spur width **402** can be equal to the spur length **363** (FIG. 6). For example, the spur width **402** and the spur length **363** (FIG. 6) can be greater than the spur height **400**. It is also contemplated in another example that the spur length **363** (FIG. 6) is greater than the spur height **400**, the spur width **402**, or both.

[0104] A third surface **406** is defined by an interior portion of the third arm **356**. A length, illustrated as a third arm length **408**, of the third arm **356** is measured from the second corner **372**

to the third corner **374**. By way of non-limiting example, the third arm length **408** can be measured along the third surface **406**.

[0105] As illustrated, by way of non-limiting example, the third arm length **408** is greater than the first arm length **392** and the second arm length **398**. It is contemplated that, for example, the third arm length **408** can be two to ten times longer than the first arm length **392**. For example, the third arm length **408** can be six to eight times the first arm length **392**. It is also contemplated that the third arm length **408** can be two to eight times the second arm length **398**. For example, the third arm length **408** can be four to six times the second arm length **398**.

[0106] The third arm length **408** can be determined by a first width of the first piece of the two adjacent pieces to be held together by the removeable clip **314**. By way of non-limiting example, the third arm length **408** can be within 10% or less of the first width of the first piece. That is, the third arm length **408** can be within 10% or less of a finished or dressed first width of the first piece that can be, for example, 0.25, 0.5, 0.75, 1.0, 1.5, 3.5, 5.5, 7.25, 9.25, or 11.25 inches. For example, the third arm length **408** can be within 5% or less of the first width of the first piece. That is, the third arm length **408** can be within 5% or less of a finished or dressed first width of the first piece that can be, for example, 0.25, 0.5, 0.75, 1.0, 1.5, 3.5, 5.5, 7.25, 9.25, or 11.25 inches.

[0107] A fourth surface **410** is defined by an interior portion of the fourth arm **358**. A length, illustrated as a fourth arm length **412**, of the fourth arm **358** is measured from the third corner **374** to the fourth corner **376**. By way of non-limiting example, the fourth arm length **412** can be measured along the fourth surface **410**.

[0108] As illustrated, by way of non-limiting example, the fourth arm length **412** is greater than the third arm length **408**, the second arm length **398**, and the first arm length **392**. Although this need not be the case. It is contemplated that, for example, the fourth arm length **412** can be two to twenty five times longer than the first arm length **392**. By way of non-limiting example, the fourth arm length **412** can be six to ten times longer than the first arm length **392**.

[0109] The fourth arm length **412** can be determined by a second width of a second piece of the two adjacent pieces to be held together by the removeable clip **314**. By way of non-limiting example, the fourth arm length **412** can be within 10% or less of the second width of the second piece plus the first thickness of the first piece. That is, fourth arm length **412** can be within 10% or less of a finished or dressed second width of the second piece that can be, for example, 0.25, 0.5, 0.75, 1.0, 1.5, 3.5, 5.5, 7.25, 9.25, or 11.25 inches added to the first thickness of the first piece that can be, for example, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 1.0, 1.125, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5, 2.75, 3.0, 3.25, or 3.5 inches. By way of further non-limiting example, the fourth arm length **412** can be within 5% or less of the second width of the second piece plus the first thickness of the first piece. That is, fourth arm length **412** can be within 5% or less of a finished or dressed second width of the second piece that can be, for example, 0.25, 0.5, 0.75, 1.0, 1.5, 3.5, 5.5, 7.25, 9.25, or 11.25 inches added to the first thickness of the first piece that can be, for example, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 1.0, 1.125, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5, 2.75, 3.0, 3.25, or 3.5 inches.

[0110] A fifth surface **414** is defined by an interior portion of the fifth arm **360**. A length, illustrated as a fifth arm length **416**, of the fifth arm **360** is measured from the fourth corner **376** to a farthest extent of the second distal end **366** of the fifth arm **360**. A fitted length **417** is measured between the fourth corner **376** to a sixth surface **420** of the sixth arm **362**. The fitted length **417** can be measured along the fifth surface **414**.

[0111] By way of non-limiting example, the fitted length **417** can be equal to the first arm length **392**. Alternatively, the fitted length **417** can be between 50% and 250% of the first arm length **392**. The second subset of spurs **369b** can be located at the fifth surface **414** between the fourth arm **358** and the sixth arm **362**. The second subset of spurs **369b** can include any number of spurs. The second subset of spurs **369b** can have equal dimensions to the first subset of spurs **369a**. However, it is contemplated that one or more spurs of the second subset of spurs **369b** can have a different spur width, spur height, or spur length. It is further contemplated that one or more spurs of the

second subset of spurs **369b** could have a different cross sectional shape than one or more of another spurs of the second subset of spurs **369b** or one or more spurs of the first subset of spurs **369a**. While illustrated as triangular prisms, it will be understood that this is by way of non-limiting example only and it is contemplated that any number of the set of spurs **369** can have any suitable shape, geometry, size, surface textures, or profile. Further still, by way of further non-limiting example, subsets of the set of spurs **369** can have different shapes, geometries, sizes, surface textures, or profiles. By way of additional non-limiting examples, one or more spurs of the set of spurs **369** can be a triangular pyramid, rectangular pyramid, or any regular or irregular pyramid. It is further contemplated that at least one spur of the set of spurs can have a cross section that is a rectangle, triangle, semi-circle, regular polygon, or irregular polygon.

[0112] The fitted length **417** can be determined by a second thickness of the second piece of the two adjacent pieces to be held together by the removeable clip **314**. By way of non-limiting example, the fitted length **417** can be within 10% or less of the second thickness. That is, fitted length **417** can be within 10% or less of a finished or dressed second thickness of the second piece that can be, for example, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 1.0, 1.125, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5, 2.75, 3.0, 3.25, or 3.5 inches. By way of further non-limiting example, the fitted length **417** can be within 5% or less of the second thickness. That is, fitted length **417** can be within 5% or less of a finished or dressed second thickness of the second piece that can be, for example, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 1.0, 1.125, 1.25, 1.5, 1.75, 2.0, 2.25, 2.5, 2.75, 3.0, 3.25, or 3.5 inches.

[0113] The sixth surface **420** is defined by an interior portion of the sixth arm **362**. A length, illustrated as a sixth arm length **422**, of the sixth arm **362** is measured from the fifth surface **414** to a free end **424**. The sixth arm length **422** is greater than the spur height **400**.

[0114] The sixth arm length **422** is in a range of 5% to 100% of the first arm length **392**. For example, the sixth arm length **422** can be in a range of 5% to 90%, 20% to 80%, 10% to 40%, 40% to 80% or 60% to 80% of the first arm length **392**.

[0115] The sixth arm **362** can include a curved portion **426**. The curved portion **426** can be adjacent or define apportion of the free end **124**. The curved portion **426** has a radius of curvature greater than zero.

[0116] An inset length **428** is defined as the length between the second distal end **366** of the fifth arm **360** and an outer portion **430** of the sixth arm **362**.

[0117] A grip **432** is defined by a portion of the fifth arm **360** between the outer portion **430** of the sixth arm **362** and the second distal end **366**. A grip length **433** can be measured from an inflection **435** of the fifth arm **360** to the distal end **366**. A grip angle **437** can be measured between the outer portion **430** of the sixth arm **362** and a grip surface **439** of the grip **432**. It is contemplated that the grip length **433** or grip angle **437** can be suitably sized to accommodate at least a portion of a finger, thumb, or hand of a user.

[0118] A first channel **434** is defined by the first arm **352**, the second arm **354**, and the third arm **356**. A second channel **436** is defined by the fourth arm **358**, the fifth arm **360**, and the sixth arm **362**.

[0119] FIG. 8 further illustrates the removable clip **314** in a mounted position or assembled with a first piece **440** and a second piece **442**. The first piece **440** and the second piece **442** can, for example, be lumber approximately 2 inches by 4 inches as sawed and usually 1.5 inches by 3.5 inches when dressed. By way of further non-limiting example, the first piece **440** or the second piece can be lumber approximately 1 inch by 2 inches, 2 inches by 6 inches, 4 inches by 4 inches, or 2 inches by 8 inches, as sawed. Further, the first piece **440** or the second piece **442** can include or be a panel, a plywood sheet, or a portion of a plywood sheet having a thickness, for example, of 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 1.125, or 1.25 inches. While illustrated as lumber, the first piece **440** or the second piece **442** can have any dimensions and be made entirely from or include portions of that are plastic, wood, or metal. That is, the removable clip **314** is coupling adjacent frames **32** (FIG. 1) illustrated as a first piece **440** and a second piece **442**, adjacent to the first piece

**440.** The first channel **434** receives the first piece **440**. In other words, when the removable clip **314** is coupling the first piece **440** and the second piece **442**, the first surface **390**, the second surface **396** and the third surface **406** are adjacent portions the first piece **440**.

[0120] The second channel **436** receives the second piece **442**. In other words, when the removable clip **314** is coupling the first piece **440** and the second piece **442**, the fourth surface **410**, the fifth surface **414**, and the sixth surface **420** are adjacent, facing, and/or in contact with portions the second piece **442**.

[0121] The first piece **440** includes a piece distal end **444** that is received by the first channel **434**. When the removable clip **314** is fastened to the first piece **440** and the second piece **442**, the first subset of spurs **369a** can pierce, detent, or extend into the first piece **440** past the piece distal end **444**. Additionally, or alternatively, the pieced distal end **444** contacts the second surface **396**.

Alternatively, the first subset of spurs **369a** can be in contact the distal end **444** without piercing. [0122] While illustrated as maintaining shape and size, it is contemplated that one or more spurs of the first subset of spurs **369a** and/or the second set of spurs **369b** can be mailable, That is, the at least one or more of the spurs of the first subset of spurs **369a** and/or the second set of spurs **369b** can bend, flex, or otherwise deform when the first piece **440** and the second piece **442** are fastened by the removable clip **314**.

[0123] A third wall thickness **446** can be measured from the third surface **406** to an outer surface **448** of the third arm **356**. Similarly, a fourth wall thickness **450** can be measured from the fourth surface **410** to an outer surface **452** of the fourth arm **358**. A third corner thickness **456** can be measured from the intersection of the third surface **406** and the fourth surface **410** to an outer surface **458** of the third corner **374**. The third corner thickness **456** can be greater than the third wall thickness **446** and the fourth wall thickness **450**. That is, the body **350** can included thickened portions at the third corner **374**. Further, as illustrated by way of example, the body **350** can included thickened portions at the first corner **370**, the second corner **372**, the third corner **374**, and the fourth corner **376**. It is contemplated that any or all of the first corner **370**, the second corner **372**, the third corner **374**, or the fourth corner **376** can include additional material or thickened portions. The thickened portions can be formed as part of the monolithic body **350**. Alternatively, the additional material or thickened portions can be coupled to the body **350** at any location.

[0124] A first radius of curvature **461** can be measured at the first corner **370** at the intersection of an outer surface of the first arm **352** and an outer surface the second arm **354**. The first radius of curvature **461** can be in a range of 25% to 100% of the spur width **402** or in a range from 1% to 25% of the first arm length **392** (FIG. 7).

[0125] A second radius of curvature **463** can be measured at the second corner **372** at the intersection of an outer surface of the second arm **354** and the outer surface **448** of the third arm **356**. The second radius of curvature **463** can be equal to the first radius of curvature **461**. Alternatively, the second radius of curvature **463** can be greater than or less than the first radius of curvature **461**.

[0126] A third radius of curvature **465** can be measured at the third corner **374** at the intersection of the outer surface **448** of the third arm **356** and the outer surface **452** of the fourth arm **358**. The third radius of curvature **465** can be greater than the first radius of curvature **461**, the second radius of curvature **463**, or both the first radius of curvature **461** and the second radius of curvature **463**. The third radius of curvature **465** can be in a range of 50% to 150% of the spur width **402** or in a range of 10% to 80% of the first arm length **392** (FIG. 7). For example, the third radius of curvature **465** can be in a range of 15% to 30% of the first arm length **392** (FIG. 7).

[0127] A fourth radius of curvature **467** can be measured at the fourth corner **376** at the intersection of the outer surface **452** of the fourth arm **358** and an outer surface of the fifth arm **360**. The fourth radius of curvature **467** can be equal to the first radius of curvature **461**, the second radius of curvature **463**, or both the first radius of curvature **461** and the second radius of curvature **463**. Alternatively, the fourth radius of curvature **467** can be different from the first radius of curvature

**461**, the second radius of curvature **463**, or both the first radius of curvature **461** and the second radius of curvature **463**.

[0128] Similar to the finger grip **132** (FIG. 3), the grip **432** can be used to couple or uncouple the removable clip **314** from the first piece **440** and the second piece **442**. For example, when the removable clip **314** fastens the first piece **440** and the second piece **442**, force can be applied to at least the grip **432** to uncouple or unfasten the removable clip **314** from first piece **440** and/or the second piece **442**.

[0129] While the invention has been specifically described in connection with certain specific embodiments thereof, it is to be understood that this is by way of illustration and not of limitation. Reasonable variation and modification are possible with the scope of the foregoing disclosure and drawings without departing from the spirit of the invention which, is defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

[0130] Further aspects of the disclosure are provided by the subject matter of the following clauses:

[0131] A removable clip for fastening together a first piece adjacent to a second piece to form a portion of a crate, the removable clip comprising a body, the body comprising a first arm, a second arm coupled to the first arm, defining a first corner, a third arm coupled to the second arm, defining a second corner, a fourth arm coupled to the third arm, defining a third corner, a fifth arm, having a first distal end and a second distal end, wherein the first distal end of the fifth arm is coupled to the fourth arm, defining a fourth corner, and a sixth arm provided on the fifth arm between the first distal end and the second distal end, wherein the first arm, the second arm, and the third arm define a first channel that receives the first piece, and wherein the fourth arm, the fifth arm, and the sixth arm define a second channel that receives the second piece.

[0132] The removable clip of any preceding clause, wherein an inset length is defined as a length between the second distal end of the fifth arm and an outer portion of the sixth arm.

[0133] The removable clip of any preceding clause, wherein a portion of the fifth arm between the second distal end of the fifth arm and the outer portion of the sixth arm defines a finger grip or grip.

[0134] The removable clip of any preceding clause, wherein a length of the first arm is measured from a first free end to the first corner and a length of the fifth arm is measured from the first distal end to the second distal end of the fifth arm, wherein the length of the fifth arm is between 50% and 200% of the length of the first arm.

[0135] The removable clip of any preceding clause, wherein a length of the sixth arm is measured from a second free end to the fifth arm, wherein the length of the sixth arm is between 5% and 100% of the length of the fifth arm.

[0136] The removable clip of any preceding clause, wherein the length of the sixth arm is between 10% and 50% of the length of the fifth arm.

[0137] The removable clip of any preceding clause, wherein the second arm includes a set of spurs between the first corner and the second corner.

[0138] The removable clip of any preceding clause, wherein the set of spurs extend a spur height from a second surface defined by the second arm, wherein the spur height is less than 15% of a length of the second arm measured from the first corner to the second corner.

[0139] The removable clip of any preceding clause, wherein the first arm, the second arm, the third arm, the fourth arm, and the fifth arm form a G-shaped clip.

[0140] The removable clip of any preceding clause, wherein the body comprises thickened portions at the first corner, the second corner, the third corner, and the fourth corner.

[0141] The removable clip of any preceding clause, wherein each of the first corner, the second corner, the third corner, and the fourth corner define an angle between 10 degrees and 110 degrees.

[0142] The removable clip of any preceding clause, wherein the angle defined by the third corner is in a range from 20 degrees to less than 90 degrees.

[0143] The removable clip of any preceding clause, wherein the body is a unitary monolithic body.

[0144] The removable clip of any preceding clause, wherein the body includes a material that is one or more of a polymer, a natural fiber, or a metal.

[0145] A removable clip for fastening together a first piece adjacent to a second piece to form a portion of a crate, the removable clip comprising a body comprising a first arm, defining a first surface, a second arm defining a second surface, wherein the second arm is coupled to the first arm, a third arm defining a third surface, wherein the third arm is coupled to the second arm, a fourth arm defining a fourth surface, wherein the fourth arm is coupled to the third arm, a fifth arm, defining a fifth surface, wherein the fifth arm has a first distal end and a second distal end, wherein the first distal end of the fifth arm is coupled to the fourth arm, and a sixth arm, defining a spur surface, wherein the sixth arm is provided on the fifth arm between the first distal end and the second distal end, wherein when the body is clipped to the first piece and the second piece, the first surface, the second surface and the third surface are adjacent portions the first piece, and wherein the fourth surface the fifth surface, and the spur surface are adjacent portions the second piece.

[0146] The removable clip of any preceding clause, wherein the body is a unitary body.

[0147] The removable clip of any preceding clause, wherein the body further comprises a finger grip or grip having an inset length defined between the second distal end of the fifth arm and an outer portion of the sixth arm.

[0148] The removable clip of any preceding clause, wherein the sixth arm includes a free end, and wherein a portion of the free end includes a curved portion.

[0149] The removable clip of any preceding clause, wherein a sixth arm length is measured from the free end to the fifth arm, wherein the sixth arm length is between 10% and 40% of a fifth arm length measured from the first distal end to the second distal end.

[0150] The removable clip of any preceding clause, wherein the body further comprises at least one spur extending a spur height from the second arm and located between a first corner and a second corner, wherein the spur height is less than the sixth arm length.

[0151] The removable clip of any preceding clause, wherein a sixth arm length is measured from the free end to the fifth arm, wherein the sixth arm length is between 20% and 80% of a fifth arm length measured from the first distal end to the second distal end.

[0152] The removable clip of any preceding clause, wherein the angle defined by the third corner is less than 90 degrees.

[0153] The removable clip of any preceding clause, wherein the second arm includes a set of spurs located between the first corner and the second corner, between the fourth corner and the sixth arm, or a combination thereof.

[0154] The removable clip of any preceding clause, wherein at least one spur of the set of spurs extend a spur height from a second surface defined by the second arm or a fifth surface defined by the fifth arm, wherein the spur height is less than 150% of a length of the second arm measured from the first corner to the second corner.

[0155] The removable clip of claim 2, wherein a portion of the fifth arm between the second distal end of the fifth arm and the outer portion of the sixth arm defines a grip length.

[0156] The removable clip of any preceding clause, wherein the set of spurs includes a first subset of spurs extending from a second surface defined by the second arm, and a second subset of spurs extending from a fifth surface defined by the fifth arm.

[0157] The removable clip of any preceding clause, wherein the set of spurs includes at least one spur with a triangular cross section.

[0158] The removable clip of any preceding clause, wherein the set of spurs includes at least one spur shaped as a triangular prism.

## Claims

- 1.** A removable clip for fastening together a first piece adjacent to a second piece to form a portion of a crate, the removable clip comprising: a body comprising: a first arm; a second arm coupled to the first arm, defining a first corner; a third arm coupled to the second arm, defining a second corner; a fourth arm coupled to the third arm, defining a third corner; a fifth arm, having a first distal end and a second distal end, wherein the first distal end of the fifth arm is coupled to the fourth arm, defining a fourth corner; and a sixth arm provided on the fifth arm between the first distal end and the second distal end; wherein the first arm, the second arm, and the third arm define a first channel that receives the first piece, and wherein the fourth arm, the fifth arm, and the sixth arm define a second channel that receives the second piece.
- 2.** The removable clip of claim 1, wherein an inset length is defined as a length between the second distal end of the fifth arm and an outer portion of the sixth arm.
- 3.** The removable clip of claim 2, wherein a portion of the fifth arm between the second distal end of the fifth arm and the outer portion of the sixth arm defines a grip length.
- 4.** The removable clip of claim 1, wherein a length of the first arm is measured from a first free end to the first corner and a length of the fifth arm is measured from the first distal end to the second distal end of the fifth arm, wherein the length of the fifth arm is between 50% and 200% of the length of the first arm.
- 5.** The removable clip of claim 4, wherein a length of the sixth arm is measured from a second free end to the fifth arm, wherein the length of the sixth arm is between 5% and 100% of the length of the fifth arm.
- 6.** The removable clip of claim 5, wherein the length of the sixth arm is between 10% and 50% of the length of the fifth arm.
- 7.** The removable clip of claim 1, wherein the second arm includes a set of spurs located between the first corner and the second corner, between the fourth corner and the sixth arm, or a combination thereof.
- 8.** The removable clip of claim 7, wherein at least one spur of the set of spurs extend a spur height from a second surface defined by the second arm or a fifth surface defined by the fifth arm, wherein the spur height is less than 15% of a length of the second arm measured from the first corner to the second corner.
- 9.** The removable clip of claim 1, wherein the first arm, the second arm, the third arm, the fourth arm, and the fifth arm form a G-shaped clip.
- 10.** The removable clip of claim 1, wherein the body comprises thickened portions at the first corner, the second corner, the third corner, and the fourth corner.
- 11.** The removable clip of claim 1, wherein each of the first corner, the second corner, the third corner, and the fourth corner define an angle between 10 degrees and 110 degrees.
- 12.** The removable clip of claim 11, wherein the angle defined by the third corner is less than 90 degrees.
- 13.** The removable clip of claim 1, wherein the body is a unitary monolithic body.
- 14.** The removable clip of claim 1, wherein the body includes a material that is one or more of a polymer, a natural fiber, or a metal.
- 15.** A removable clip for fastening together a first piece adjacent to a second piece to form a portion of a crate, the removable clip comprising: a body comprising: a first arm, defining a first surface; a second arm defining a second surface, wherein the second arm is coupled to the first arm; a third arm defining a third surface, wherein the third arm is coupled to the second arm; a fourth arm defining a fourth surface, wherein the fourth arm is coupled to the third arm; a fifth arm, defining a fifth surface, wherein the fifth arm has a first distal end and a second distal end, wherein the first distal end of the fifth arm is coupled to the fourth arm; and a sixth arm, defining a spur surface, wherein the sixth arm is provided on the fifth arm between the first distal end and the second distal end; wherein when the body is clipped to the first piece and the second piece, the first surface, the

second surface and the third surface are adjacent portions the first piece, and wherein the fourth surface the fifth surface, and the spur surface are adjacent portions the second piece.

**16.** The removable clip of claim 15, wherein the body is a unitary body.

**17.** The removable clip of claim 15, wherein the body further comprises a grip having an inset length defined between the second distal end of the fifth arm and an outer portion of the sixth arm.

**18.** The removable clip of claim 15, wherein the sixth arm includes a free end, and wherein a portion of the free end includes a curved portion.

**19.** The removable clip of claim 18, wherein a sixth arm length is measured from the free end to the fifth arm, wherein the sixth arm length is between 20% and 80% of a fifth arm length measured from the first distal end to the second distal end.

**20.** The removable clip of claim 19, wherein the body further comprises at least one spur extending a spur height from the second arm and located between a first corner and a second corner, wherein the spur height is less than the sixth arm length.

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